

# 6. AC Steady-State Analysis

## Introduction to Electric Circuits

### Supplemental Instruction

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*This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.*

## 1 Introduction

We are now beginning our study of AC circuits! In AC, or Alternating Current, both voltage and current are sinusoidal functions of time, meaning they alternate back and forth between positive and negative values. Working with sinusoidal functions rather than simple numbers might appear difficult at first, but there are a number of mathematical tools and tricks we can use to simplify our analysis. Here I've attempted to condense these down into a minimal "cheat sheet" of what you need to know.

## 2 The Sinusoidal Function

Sinusoidal voltages and currents are written as

$$m(t) = M \cos(\omega t + \alpha)$$

where  $M$  is the amplitude (sometimes called the peak amplitude),  $\omega$  is the angular frequency (in rad/s), and  $\alpha$  is the phase angle.

The angular frequency is related to the frequency by the equation

$$\omega = 2\pi f$$

Where  $\omega$  is in rad/s and  $f$  is in Hz. The period  $T$  of a sinusoidal function, in seconds, is

$$T = \frac{1}{f}$$

Note that the  $\sin()$  function is not used (by convention). If a sin function was written in the standard form, it would be as  $\cos(x - 90^\circ) = \sin(x)$ .

### Leading and Lagging Signals

Let's say we have two sinusoidal signals of the same frequency,  $v_1(t) = V_{1m} \cos(\omega t + \alpha_1)$  and  $v_2(t) = V_{2m} \cos(\omega t + \alpha_2)$ . If  $\alpha_1 > \alpha_2$ , we say that  $v_1$  **leads**  $v_2$ , and at the same time,  $v_2$  **lags**  $v_1$ . Similarly, if  $\alpha_2 > \alpha_1$ ,  $v_2$  leads  $v_1$ , and  $v_1$  lags  $v_2$ .

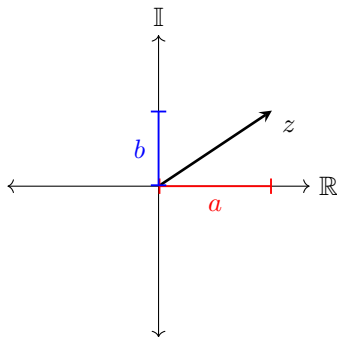
## 3 Complex Number Review

### 3.1 Notation

Recall that the imaginary unit is  $i = \sqrt{-1}$ . However, for circuits, we use  $j$  instead of  $i$  so as not to confuse it with current. A complex number can be written in **rectangular form** as

$$\mathbf{Z} = a + bj$$

Where  $a = \text{Re}\{z\}$  is the real part, and  $b = \text{Im}\{z\}$  is the imaginary part. We can also draw a complex number as a vector in the complex plane:



Another way to write complex numbers is **polar form**:

$$\mathbf{Z} = Z_m \angle \theta$$

Where  $Z_m$  is the magnitude and  $\theta$  is the angle. The  $\angle$  symbol can be pronounced "angle". An equivalent notation for polar form is

$$\mathbf{Z} = Z_m e^{j\theta}$$

We can convert between polar and rectangular forms using

$$Z_m = \sqrt{a^2 + b^2}, \quad \tan(\theta) = \frac{b}{a}$$

Many scientific and graphing calculators have built-in functions that can do these conversions.

### 3.2 Complex Multiplication and Division

Let's say we have two complex numbers,  $\mathbf{X} = X_m \angle \alpha$  and  $\mathbf{Y} = Y_m \angle \beta$ . We can write their product as

$$\mathbf{XY} = X_m e^{j\alpha} \cdot Y_m e^{j\beta} = X_m \cdot Y_m e^{j\alpha+j\beta} = X_m \cdot Y_m e^{j(\alpha+\beta)} = X_m \cdot Y_m \angle (\alpha + \beta)$$

We can also divide complex numbers:

$$\frac{\mathbf{X}}{\mathbf{Y}} = \frac{X_m e^{j\alpha}}{Y_m e^{j\beta}} = \frac{X_m}{Y_m} e^{j\alpha-j\beta} = \frac{X_m}{Y_m} \angle (\alpha - \beta)$$

### 3.3 A Note on the Complex Unit

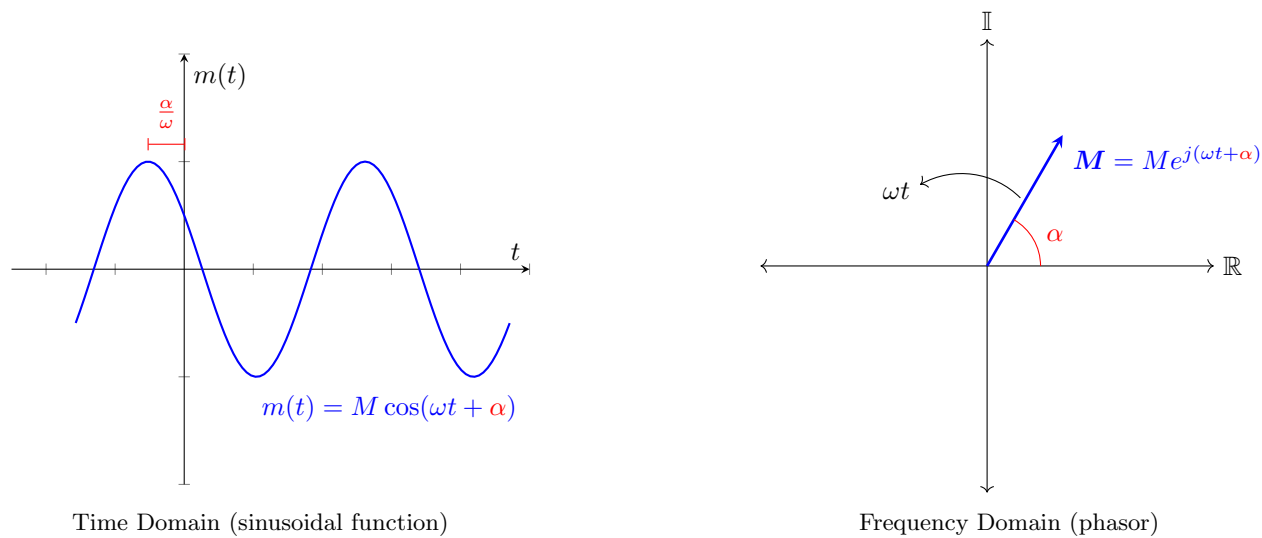
Note that, using polar form,  $j = 1 \angle 90^\circ$ . Also, it can be shown using the division formula above that  $\frac{1}{j} = -j$ .

## 4 The Phasor (yes, they're really called that, it's awesome)

Using math, it is possible to write a sinusoidal function in the *time domain* as a **phasor** in the *frequency domain*:

$$m(t) = M \cos(\omega t + \alpha) \Rightarrow \mathbf{M}(t) = M e^{j(\omega t + \alpha)}$$

$\mathbf{M}(t)$  is a vector in the complex plane that rotates counterclockwise over time, with magnitude  $M$  and angle  $(\omega t + \alpha)$ . Below is a graphical representation of a sinusoidal function and its phasor representation. We usually draw phasors at time  $t = 0$ , so the phase angle  $\alpha$  is the angle between the phasor and the positive real axis.



### Phasor Notation

In this class we only work with circuits in which every time-varying value is a sinusoid of the same frequency. Therefore, the  $\omega t$  term is present in every phasor, so we usually abbreviate by omitting it and writing a phasor as simply

$$m(t) = M \cos(\omega t + \alpha) \Rightarrow \mathbf{M} = M e^{j\alpha}$$

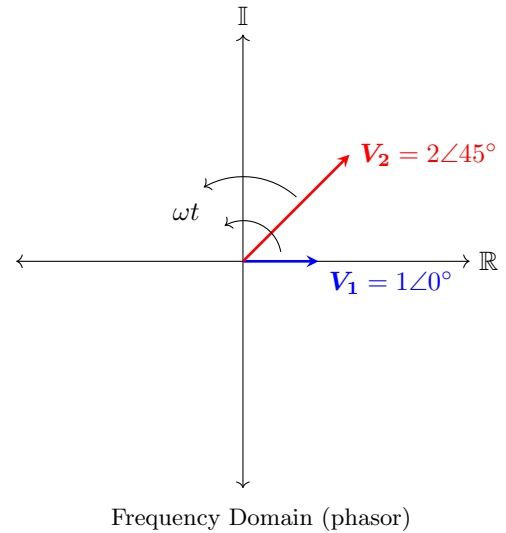
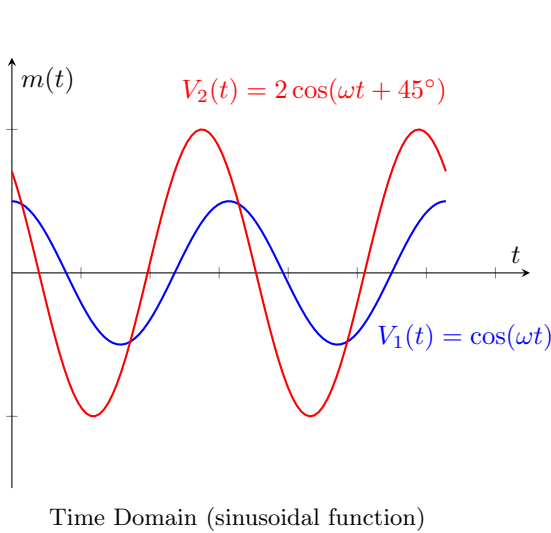
Or, using abbreviated polar form notation:

$$\mathbf{M} = M \angle \alpha$$

Phasors can also be written in rectangular form:

$$\mathbf{M} = a + bj, \quad a = M \cos(\alpha), \quad b = M \sin(\alpha)$$

Below is a visual representation of some different phasors using the abbreviated polar form:



## Handwriting Phasors

In typed documents such as this, phasors are written using a bold font:  $\mathbf{V}$ . However, in handwriting, a dot is written above the symbol to specify that is is a phasor:  $\dot{V}$ .

## 5 Impedance

All of the circuit analysis we've done up to this point works the same way in the frequency domain, but every current and voltage is a complex number (a phasor). The equivalent of resistance when working with phasors is impedance ( $\mathbf{Z}$ ), which is a complex number but not a phasor (it doesn't rotate over time). For any impedance  $\mathbf{Z}$ , Ohm's Law is true in the frequency domain:

$$\mathbf{V} = \mathbf{I} \cdot \mathbf{Z}$$

The unit of impedance is Ohms ( $\Omega$ ). The impedance of various components can be found as follows:

|           |  |                                      |
|-----------|--|--------------------------------------|
| Resistor  |  | $\mathbf{Z}_R = R$                   |
| Inductor  |  | $\mathbf{Z}_L = j\omega L$           |
| Capacitor |  | $\mathbf{Z}_C = \frac{1}{j\omega C}$ |

In rectangular form, impedance is written as

$$\mathbf{Z} = R + Xj$$

where  $R$  is resistance, and  $X$  is called **reactance**, and both have units of Ohms ( $\Omega$ ). Note that reactance is a real number (it does not include  $j$ ). For an inductor,  $X = \omega L$ , and for a capacitor,  $X = \frac{-1}{\omega C}$ . Note that resistors have zero reactance, inductors have positive reactance, and capacitors have negative reactance. However, sometimes capacitor reactance is written as a positive number,  $X_c = \frac{1}{\omega C}$ ; in this case the negative sign is implied.

## Admittance and Susceptance

The reciprocal of impedance is admittance,  $\mathbf{Y}$ :

$$\mathbf{Y} = \frac{1}{\mathbf{Z}}$$

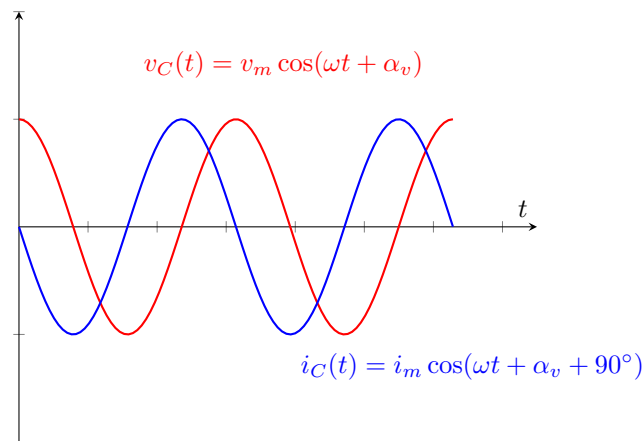
In rectangular form,

$$\mathbf{Y} = G + jB$$

Where  $G$  is conductance, and  $B$  is called **Susceptance**, and both have unites of Siemens (s).

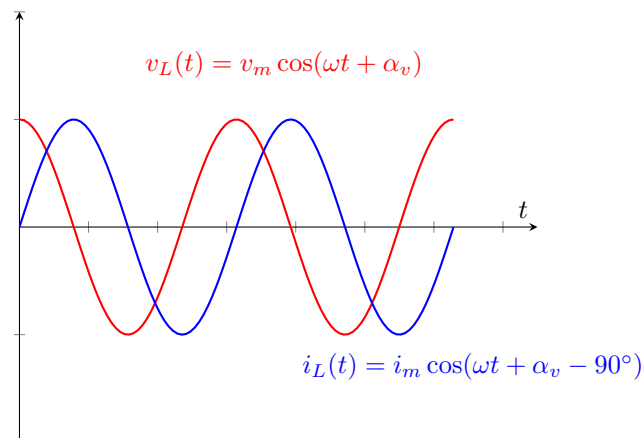
## Current and Voltage in a Capacitor

In a capacitor, current leads voltage by  $90^\circ$ :



## Current and Voltage in an Inductor

In an inductor, current lags voltage by  $90^\circ$ :



The above statements can be proven using either phasors and impedance or time domain analysis.

## 6 Appendix: Why Complex Numbers Work for Circuit Analysis

### Laplace Domain Analysis

Remember that for an inductor,  $v_L(t) = L \frac{di_L(t)}{dt}$ , and for a capacitor,  $i_C(t) = C \frac{dv_C(t)}{dt}$ , which means  $v_C(t) = \frac{1}{C} \int i_C(t)$  (ignoring initial conditions). When we take the Laplace transform, we treat differentiation as multiplication by  $s$ , and integration as division by  $s$ . Therefore, taking the Laplace transform, we get

$$V_L(s) = LsI_L(s) \quad \text{and} \quad V_C(s) = \frac{I_C(s)}{Cs}$$

If we substitute  $s = j\omega$ <sup>1</sup>, we get

$$V_L(s) = j\omega LI_L(s) \quad \text{and} \quad V_C(s) = \frac{I_C(s)}{j\omega C}$$

We can now substitute the impedances  $Z_L = j\omega L$  and  $Z_C = \frac{1}{j\omega C}$ :

$$V_L(s) = Z_L I_L(s) \quad \text{and} \quad V_C(s) = Z_C I_C(s)$$

And we have arrived at Ohm's Law in the frequency domain.

### Treating multiplication by $j\omega$ as differentiation

Another way of looking at this is as follows. If we have an inductor with some current  $i(t) = I_m \cos(\omega t + \alpha)$ , the voltage will be

$$v(t) = L \cdot \frac{di(t)}{dt} = L \cdot -I_m \omega \sin(\omega t + \alpha)$$

We can use the trig identity  $-\sin(x) = \cos(x + 90^\circ)$  to simplify:

$$v(t) = L \cdot I_m \omega \cos(\omega t + \alpha + 90^\circ)$$

We can write this as a phasor:

$$\mathbf{V} = LI_m \omega \angle(\alpha + 90^\circ)$$

Using Ohm's Law, we can equate this voltage to the current times the impedance:

$$\mathbf{V} = LI_m \omega \angle(\alpha + 90^\circ) = \mathbf{I} \mathbf{Z}_L$$

Let's write the impedance and current in polar form with an impedance magnitude  $Z_m$  and phase angle  $\theta$ .

$$\mathbf{V} = LI_m \omega \angle(\alpha + 90^\circ) = I_m \angle \alpha \cdot Z_L \angle \theta_L = I_m Z_L \angle(\alpha + \theta_L)$$

Equating the magnitude and angle:

$$LI_m \omega = I_m Z_L \Rightarrow Z_L = L\omega$$

$$\alpha + 90^\circ = \alpha + \theta_L \Rightarrow \theta_L = 90^\circ$$

Writing  $\mathbf{Z}_L$  in rectangular form,

$$\mathbf{Z}_L = j\omega L$$

We can find the impedance of a capacitor in a similar manner.

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<sup>1</sup>The reasoning behind this substitution is not something I am able to explain. I would recommend [3Blue1Brown's youtube series on the topic](#) for those who want to learn more.

## References

- M. Iordache. Class Lectures, Topic: “Electric circuits.” EEGR 2053, School of Engineering and Engineering Technology, LeTourneau University, 2025.
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